

Model-Based Machine Learning

Notes des Cours Magistraux

53LIED01 / 53LIED02
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September 26, 2026



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1 Introduction

1.1 Notations

- Observations are described by p features $\mathbf{X} = (X_1, \dots, X_p) \in E$ ($E = \mathbb{R}^p, \dots$)
- $\mathbf{X}_i = (X_{i1}, \dots, X_{ip})$ is the features for observation i ($1 \leq i \leq n$)
- $Z_i \in \{1, \dots, K\}$ is the group of observation i

1.2 The Mixture Model

Idea: each group is described by its *own* probability distribution.

$$\mathbf{X} \mid Z = k \sim f(\mathbf{x}, \theta_k) = f_k(\mathbf{x})$$

How to read it:

- $\mathbf{X}$ = feature vector describing one observation.
- $Z = k$ = the observation belongs to group/cluster k .
- $\mathbf{X} \mid Z = k$ = the features *given that* the observation belongs to group k .
- $\sim$ = “is distributed according to”.
- $f_k(\mathbf{x})$ = the probability distribution (density) for group k .
- θ_k = parameters describing the distribution for group k .

If an observation belongs to group k , its features follow the distribution f_k with parameters θ_k .

Key idea: Each group has its *own* probability distribution:

$$f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_K(\mathbf{x}).$$

Important for clustering: We observe $\mathbf{X}_i$, but Z_i is typically *unknown*. We therefore use the observed features to infer which group generated each observation.

1.3 Mixing Proportions and Latent Group Membership

The probability that an observation belongs to group k is called the **mixing proportion**:

$$p_k = P(Z = k), \quad k = 1, \dots, K.$$

The mixing proportions satisfy

$$p_k \geq 0 \quad \text{and} \quad \sum_{k=1}^K p_k = 1.$$

Thus, p_k represents the proportion (or prior probability) of observations belonging to group k .

One-hot representation of the group membership:

The group variable Z can be represented by a vector $\tilde{Z} = (\tilde{Z}_1, \dots, \tilde{Z}_K)$ such that

$$Z = k \iff \tilde{Z} = (0, \dots, 0, 1, 0, \dots, 0),$$

where the 1 is in the k -th position. Therefore,

$$\tilde{Z}_k = \begin{cases} 1 & \text{if } Z = k, \\ 0 & \text{otherwise.} \end{cases}$$

The membership vector follows a multinomial distribution with one trial:

$$\tilde{Z} \sim \mathcal{M}(1, p_1, \dots, p_K),$$

and

$$p_k = P(Z = k) = P(\tilde{Z}_k = 1).$$

Key idea: First, a group is selected according to the mixing proportions $p_1, \dots, p_K$. Then, the observation is generated from the corresponding group distribution.

1.4 Marginal Distribution of $\mathbf{X}$: Mixture Density

$\mathbf{X}$ Conditionally on belonging to group k ,

$$\mathbf{X} \mid Z = k \sim f_k(\mathbf{x}).$$

Since Z is generally unobserved, the marginal distribution of $\mathbf{X}$ is obtained by combining all group distributions:

$$f(\mathbf{x}) = \sum_{k=1}^K p_k f_k(\mathbf{x})$$

This is called the **mixture density**.

$$\text{Mixture density} = \sum \text{mixing proportion} \times \text{group density}$$

1.5 Posterior Probability of Group Membership

For an observed feature vector $\mathbf{X} = \mathbf{x}$, we want to compute the probability that it belongs to group k :

$$P(Z = k \mid \mathbf{X} = \mathbf{x}).$$

Using Bayes' theorem:

$$P(Z = k \mid \mathbf{X} = \mathbf{x}) = \frac{p_k f_k(\mathbf{x})}{\sum_{\ell=1}^K p_{\ell} f_{\ell}(\mathbf{x})} = \frac{p_k f_k(\mathbf{x})}{f(\mathbf{x})}$$

Interpretation:

- p_k : how common group k is (*prior probability*);
- $f_k(\mathbf{x})$: how well group k explains the observation $\mathbf{x}$;
- $P(Z = k \mid \mathbf{X} = \mathbf{x})$: the probability that the observation belongs to group k *after observing $\mathbf{x}$* (*posterior probability*).

Memory aid:

$$\text{Prior probability} \times \text{How well the group explains } \mathbf{x} \longrightarrow \text{Posterior probability}$$

2 Classification

2.1 Classification Rule

A **classification rule** assigns each observation $\mathbf{x} \in E$ to one of the K groups:

$$r : E \rightarrow \{1, \dots, K\}.$$

The rule is equivalent to partitioning the feature space E into K regions:

$$\Omega_k = \{\mathbf{x} \in E : r(\mathbf{x}) = k\},$$

so that

$$E = \bigcup_{k=1}^K \Omega_k, \quad \Omega_k \cap \Omega_l = \emptyset \quad (k \neq l).$$

Interpretation: Ω_k is the region of feature space that the classifier assigns to group k .

2.2 Classification Errors

If an observation truly belongs to group G_k , the probability of classifying it as group G_l ($l \neq k$) is

$$P(r(\mathbf{X}) = l \mid Z = k) = P(\mathbf{X} \in \Omega_l \mid Z = k) = \int_{\Omega_l} f_k(\mathbf{x}) d\mathbf{x}$$

The probability of misclassifying an observation from group G_k is

$$P(r(\mathbf{X}) \neq k \mid Z = k) = \sum_{l \neq k} P(r(\mathbf{X}) = l \mid Z = k)$$

The **global probability of misclassification** (global classification error) is

$$P(r(\mathbf{X}) \neq Z) = \sum_{k=1}^K p_k P(r(\mathbf{X}) \neq k \mid Z = k)$$

2.3 Cost and Risk

Let C_{kl} be the cost of classifying an observation from true group G_l as group G_k . Usually,

$$C_{kk} = 0.$$

The **conditional risk** of assigning an observation $\mathbf{x}$ to group k is the expected cost:

$$R(k | \mathbf{x}) = \sum_{l=1}^K C_{kl} P(Z = l | \mathbf{X} = \mathbf{x})$$

The **average risk** of a classification rule r is

$$R(r) = \mathbb{E}[C_{r(\mathbf{X}), Z}]$$

Interpretation:

$$\text{Risk} = \text{expected cost of classification errors}$$

2.4 Bayes Optimal Classification Rule

The **Bayes optimal rule** is the rule that minimizes the average risk:

$$r^* = \arg \min_r R(r)$$

Equivalently, for each observation $\mathbf{x}$, choose the group with the smallest conditional risk:

$$r^*(\mathbf{x}) = \arg \min_k R(k | \mathbf{x})$$

2.5 Bayes Rule with Equal Costs

If every incorrect classification has the same cost,

$$C_{kl} = \begin{cases} 0 & k = l, \\ 1 & k \neq l, \end{cases}$$

then minimizing conditional risk is equivalent to maximizing the posterior probability:

$$r^*(\mathbf{x}) = \arg \max_k P(Z = k | \mathbf{x})$$

Using Bayes' theorem,

$$P(Z = k | \mathbf{x}) = \frac{p_k f_k(\mathbf{x})}{\sum_{l=1}^K p_l f_l(\mathbf{x})},$$

so equivalently,

$$r^*(\mathbf{x}) = \arg \max_k p_k f_k(\mathbf{x})$$

2.6 Bayes Rule for Two Groups

For $K = 2$:

$$r^*(\mathbf{x}) = \begin{cases} 1 & \text{if } p_1 f_1(\mathbf{x}) > p_2 f_2(\mathbf{x}), \\ 2 & \text{otherwise.} \end{cases}$$

The decision boundary is given by

$$\boxed{p_1 f_1(\mathbf{x}) = p_2 f_2(\mathbf{x})}$$

3 Univariate vs. Multivariate Gaussian Distributions

This section compares the mathematical definitions and structural components of the 1D (univariate) and multi-dimensional (multivariate) Gaussian distributions.

3.1 Probability Density Functions (PDFs)

3.1.1 Univariate Gaussian Distribution

For a single continuous random variable x , the probability density function is defined as:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (1)$$

Where:

- μ is the scalar mean (location of the peak).
- σ^2 is the scalar variance (spread of the distribution).

3.1.2 Multivariate Gaussian Distribution

For a D -dimensional vector of random variables $\mathbf{x} = [x_1, x_2, \dots, x_D]^T$, the probability density function generalizes to:

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) \quad (2)$$

Where:

- $\boldsymbol{\mu} \in \mathbb{R}^D$ is the mean vector.
- $\boldsymbol{\Sigma} \in \mathbb{R}^{D \times D}$ is the symmetric, positive-definite covariance matrix.
- $|\boldsymbol{\Sigma}|$ denotes the determinant of the covariance matrix.
- $\boldsymbol{\Sigma}^{-1}$ is the inverse of the covariance matrix.

3.2 Mathematical Component Mapping

The table below illustrates how the components of the univariate formula scale up to handle multiple dimensions.

Table 1: Structural Translation from 1D to D-Dimensions

Component	Univariate (1D)	Multivariate (D-D)	Structural Role
Input Variable	Scalar: x	Vector: $\mathbf{x} \in \mathbb{R}^D$	Tracks multiple dimensions simultaneously.
Center / Mean	Scalar: μ	Vector: $\boldsymbol{\mu} \in \mathbb{R}^D$	Contains individual coordinates for the center.
Spread / Scale	Variance: σ^2	Matrix: $\boldsymbol{\Sigma} \in \mathbb{R}^{D \times D}$	Determinant $ \boldsymbol{\Sigma} $ scales the density volume.
Inverse Spread	Division: $\frac{1}{\sigma^2}$	Matrix Inverse: $\boldsymbol{\Sigma}^{-1}$	Matrix inversion replaces regular division.
Distance Metric	$(x - \mu)^2$	$(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$	Computes squared Mahalanobis distance.
Normalization	$\sqrt{2\pi\sigma^2}$	$\sqrt{(2\pi)^D \boldsymbol{\Sigma} }$	Scales total density volume to integrate to 1.

3.3 Key Geometric Interpretation

The exponent term in the multivariate PDF represents the *Mahalanobis distance*. Unlike the standard Euclidean distance, it accounts for variances along each axis and covariances between dimensions. This turns the perfectly circular or spherical contours of independent variables into multi-dimensional hyperellipsoids when variables correlate.

4 GMM & MLE

4.1 Gaussian Mixture Model (GMM)

A **Gaussian Mixture Model** assumes that each group has a multivariate Gaussian distribution:

$$\mathbf{X} \mid Z = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$$

Thus,

$$f_k(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k).$$

The marginal mixture density is

$$f(\mathbf{x}) = \sum_{k=1}^K p_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)$$

Each group is characterized by

$$p_k, \quad \boldsymbol{\mu}_k, \quad \Sigma_k$$

where p_k is the mixing proportion, $\boldsymbol{\mu}_k$ the mean, and Σ_k the covariance matrix.

4.2 Maximum Likelihood Estimation (MLE)

Given observations $\mathbf{x}_1, \dots, \mathbf{x}_n$, MLE chooses the parameters that make the observed data most likely:

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

where the likelihood is

$$L(\theta) = \prod_{i=1}^n f(\mathbf{x}_i; \theta).$$

4.3 Log-Likelihood

Because products are difficult to optimize, we use the logarithm:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(\mathbf{x}_i; \theta)$$

Since log is increasing,

$$\arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta).$$

4.4 Empirical Estimates

If the group memberships are known, let

$$\tilde{Z}_{ik} = \begin{cases} 1 & \text{if observation } i \text{ belongs to group } k, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik}$$

is the number of observations in group k .
The usual empirical estimates are

$$\hat{p}_k = \frac{n_k}{n}$$

$$\hat{\boldsymbol{\mu}}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} \mathbf{x}_i$$

and

$$\hat{\Sigma}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^T$$

Important: These estimates assume that the group memberships $\tilde{Z}_{ik}$ are known. In unsupervised GMM clustering, the memberships are unknown and must be estimated, which leads to methods such as Expectation-Maximization (EM).

4.5 Key Connections

$$\text{GMM} \rightarrow p_k, f_k(\mathbf{x}) \rightarrow P(Z = k \mid \mathbf{x}) \rightarrow \text{classification} \rightarrow \text{risk} \rightarrow \text{Bayes optimal rule}$$

$$\text{MLE} \rightarrow \text{estimate } p_k, \boldsymbol{\mu}_k, \Sigma_k \rightarrow \text{fit the GMM}$$

Memory aids:

- **Classification rule:** “Which group do I assign $\mathbf{x}$ to?”
- **Risk:** “How costly is my decision?”
- **Bayes rule:** “Choose the decision with the smallest expected cost.”
- **Equal costs:** “Choose the group with the highest posterior probability.”

- **GMM:** “Each group is a Gaussian with its own p_k, μ_k, Σ_k .”
- **MLE:** “Choose parameters that make the observed data most likely.”

5 Clustering and the EM Algorithm

5.1 Likelihood and Log-Likelihood in the Clustering Context

In the clustering setting, the observations

$$\underline{X} = (X_1, \dots, X_n)$$

are observed, but the group memberships

$$\underline{Z} = (Z_1, \dots, Z_n)$$

are **unobserved (latent)**.

For a Gaussian mixture model,

$$X_i \mid Z_i = k \sim \mathcal{N}(\mu_k, \Sigma_k),$$

with mixing proportions

$$P(Z_i = k) = p_k, \quad \sum_{k=1}^K p_k = 1.$$

The density of an observation x_i is therefore obtained by summing over the possible groups:

$$f(x_i; \theta) = \sum_{k=1}^K p_k f_k(x_i; \theta_k).$$

Hence, assuming independent observations, the **observed-data likelihood** is

$$L(\theta; \underline{X}) = \prod_{i=1}^n \left(\sum_{k=1}^K p_k f_k(x_i; \theta_k) \right).$$

The corresponding observed-data log-likelihood is

$$\ell(\theta; \underline{X}) = \log L(\theta; \underline{X}) = \sum_{i=1}^n \log \left(\sum_{k=1}^K p_k f_k(x_i; \theta_k) \right).$$

Why is the clustering likelihood harder? When Z_i is known, the likelihood contains products of the form

$$p_k f_k(x_i; \theta_k),$$

because we know which group generated x_i .

In clustering, however, Z_i is unknown. Therefore, for each observation x_i , we must consider all possible groups:

$$f(x_i; \theta) = \sum_{k=1}^K p_k f_k(x_i; \theta_k).$$

The logarithm is then applied to this sum:

$$\log \left(\sum_{k=1}^K p_k f_k(x_i; \theta_k) \right).$$

In general,

$$\log \left(\sum_k a_k \right) \neq \sum_k \log(a_k),$$

which makes the direct maximisation of the observed likelihood more difficult.

Important distinction. The **observed-data likelihood** uses only the observed variables:

$$L(\theta; \underline{X}) = \prod_{i=1}^n \sum_{k=1}^K p_k f_k(x_i; \theta_k).$$

The **complete-data likelihood** assumes that the latent group memberships Z are also available:

$$L_c(\theta; \underline{X}, Z) = \prod_{i=1}^n \prod_{k=1}^K [p_k f_k(x_i; \theta_k)]^{\tilde{Z}_{ik}}.$$

The difference between these two likelihoods is the key reason why EM is introduced.

5.2 MLE in the Clustering Context

The maximum likelihood estimator in clustering is defined by

$$\hat{\theta} = \arg \max_{\theta} L(\theta; \underline{X})$$

or, equivalently,

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta; \underline{X}).$$

For a Gaussian mixture model,

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^n \log \left(\sum_{k=1}^K p_k f_k(x_i; \mu_k, \Sigma_k) \right).$$

The difficulty is that the latent variables Z_i are unknown.

If the memberships were known, we could directly maximise the complete-data likelihood and obtain the empirical estimators

$$\hat{p}_k = \frac{n_k}{n},$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} x_i,$$

and

$$\hat{\Sigma}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T.$$

However, in clustering the $\tilde{Z}_{ik}$ are unknown.
Therefore, we cannot directly use these formulas.

The problem is that the quantities needed to estimate the parameters are themselves unknown.

The Expectation-Maximisation algorithm solves this problem by alternating between estimating the latent memberships and updating the model parameters.

5.3 From the Observed Likelihood to the Completed Likelihood

The central idea of EM is to work with the **complete-data likelihood**, because it is easier to maximise.

The complete-data likelihood is

$$L_c(\theta; \underline{X}, Z) = P(\underline{X}, Z; \theta).$$

Using the one-hot representation $\tilde{Z}_{ik}$,

$$L_c(\theta; \underline{X}, Z) = \prod_{i=1}^n \prod_{k=1}^K [p_k f_k(x_i; \theta_k)]^{\tilde{Z}_{ik}}.$$

Its logarithm is

$$\ell_c(\theta; \underline{X}, Z) = \sum_{i=1}^n \sum_{k=1}^K \tilde{Z}_{ik} [\log p_k + \log f_k(x_i; \theta_k)].$$

This is much easier to handle because there is no logarithm of a sum.
However, Z is unknown.

Therefore, instead of maximising

$$\ell_c(\theta; \underline{X}, Z)$$

directly, EM uses its conditional expectation given the observed data.

For a current parameter estimate $\theta^{(t)}$, define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Z \mid X, \theta^{(t)}} [\ell_c(\theta; X, Z)].$$

The algorithm then alternates between:

E-step: estimate the latent variables

and

M-step: maximise the expected complete-data log-likelihood.

5.4 The EM Algorithm

Let

$$\theta^{(t)}$$

be the current estimate of the model parameters.

5.4.1 E-Step: Expectation

The goal is to compute the conditional distribution of the latent variables given the observations and the current parameters.

For each observation i and group k , define

$$\tau_{ik}^{(t)} = P(Z_i = k \mid X_i = x_i, \theta^{(t)}).$$

Using Bayes' rule,

$$\tau_{ik}^{(t)} = \frac{p_k^{(t)} f_k(x_i; \theta_k^{(t)})}{\sum_{\ell=1}^K p_\ell^{(t)} f_\ell(x_i; \theta_\ell^{(t)})}.$$

These quantities are called the **responsibilities**.

They satisfy

$$0 \leq \tau_{ik}^{(t)} \leq 1$$

and

$$\sum_{k=1}^K \tau_{ik}^{(t)} = 1.$$

Intuitively,

$$\tau_{ik}^{(t)} = \text{probability that observation } i \text{ belongs to group } k$$

according to the current model.

The E-step can therefore be viewed as replacing the unknown hard membership indicators $\tilde{Z}_{ik}$ by their conditional expectations:

$$\mathbb{E}[\tilde{Z}_{ik} \mid X, \theta^{(t)}] = \tau_{ik}^{(t)}.$$

Hence,

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K \tau_{ik}^{(t)} [\log p_k + \log f_k(x_i; \theta_k)].$$

5.4.2 M-Step: Maximisation

We now maximise the expected complete-data log-likelihood:

$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta \mid \theta^{(t)}).$$

For a Gaussian mixture, this gives

$$p_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \tau_{ik}^{(t)},$$

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)} x_i}{\sum_{i=1}^n \tau_{ik}^{(t)}},$$

and

$$\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)} (x_i - \mu_k^{(t+1)})(x_i - \mu_k^{(t+1)})^T}{\sum_{i=1}^n \tau_{ik}^{(t)}}.$$

Thus, the observations are no longer assigned completely to one group. Instead, each observation contributes to every group with weight $\tau_{ik}^{(t)}$.

5.4.3 Complete EM Iteration

The EM algorithm is therefore

$$\boxed{\theta^{(t)} \xrightarrow{\text{E-step}} \tau_{ik}^{(t)} \xrightarrow{\text{M-step}} \theta^{(t+1)}}.$$

More explicitly:

1. Choose an initial parameter value $\theta^{(0)}$.
2. **E-step:** compute the responsibilities

$$\tau_{ik}^{(t)} = P(Z_i = k \mid X_i = x_i, \theta^{(t)}).$$

3. **M-step:** maximise

$$Q(\theta \mid \theta^{(t)})$$

and obtain $\theta^{(t+1)}$.

4. Repeat the E-step and M-step until convergence.

A common stopping criterion is based on the observed log-likelihood:

$$\left| \ell(\theta^{(t+1)}) - \ell(\theta^{(t)}) \right| < \varepsilon.$$

5.5 Proof of EM Monotonicity and Convergence

We now prove the fundamental property of EM:

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})}.$$

Thus, the observed log-likelihood does not decrease from one EM iteration to the next.

Step 1: The complete-data distribution

Let

$$p_\theta(x, z)$$

denote the joint density of the observed and latent variables.

Then

$$p_\theta(x, z) = p_\theta(z \mid x) p_\theta(x).$$

Taking logarithms gives

$$\log p_\theta(x, z) = \log p_\theta(z \mid x) + \log p_\theta(x).$$

For fixed observed data x , define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} [\log p_\theta(X, Z) \mid X = x].$$

Therefore,

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} [\log p_\theta(Z \mid X) \mid X = x] + \log p_\theta(x).$$

Similarly,

$$Q(\theta^{(t)} \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} [\log p_{\theta^{(t)}}(Z \mid X) \mid X = x] + \log p_{\theta^{(t)}}(x).$$

Step 2: Subtract the two expressions

Subtracting gives

$$\begin{aligned} & Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}) \\ &= \mathbb{E}_{\theta^{(t)}} \left[\log \frac{p_\theta(Z \mid X)}{p_{\theta^{(t)}}(Z \mid X)} \mid X = x \right] + \ell(\theta; x) - \ell(\theta^{(t)}; x). \end{aligned}$$

By definition of the Kullback–Leibler divergence,

$$\begin{aligned} & \text{KL}(p_{\theta^{(t)}}(Z \mid X) \parallel p_\theta(Z \mid X)) \\ &= \mathbb{E}_{\theta^{(t)}} \left[\log \frac{p_{\theta^{(t)}}(Z \mid X)}{p_\theta(Z \mid X)} \mid X = x \right]. \end{aligned}$$

Hence,

$$\mathbb{E}_{\theta^{(t)}} \left[\log \frac{p_\theta(Z \mid X)}{p_{\theta^{(t)}}(Z \mid X)} \mid X = x \right] = -\text{KL}(p_{\theta^{(t)}}(Z \mid X) \parallel p_\theta(Z \mid X)).$$

Therefore,

$$\boxed{\ell(\theta; x) - \ell(\theta^{(t)}; x) = Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}) + \text{KL}(p_{\theta^{(t)}}(Z \mid X) \parallel p_\theta(Z \mid X))}.$$

Step 3: Use the non-negativity of KL divergence

Recall that

$$\text{KL}(P\|Q) \geq 0.$$

Thus,

$$\ell(\theta; x) - \ell(\theta^{(t)}; x) \geq Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}).$$

In the M-step, $\theta^{(t+1)}$ is chosen such that

$$Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)}).$$

Therefore,

$$\ell(\theta^{(t+1)}; x) - \ell(\theta^{(t)}; x) \geq 0.$$

Hence,

$$\boxed{\ell(\theta^{(t+1)}; x) \geq \ell(\theta^{(t)}; x)}.$$

This proves that EM produces a non-decreasing sequence of observed log-likelihood values.

Step 4: Consequence for the likelihood sequence

Define

$$\ell_t = \ell(\theta^{(t)}; x).$$

From the previous result,

$$\ell_{t+1} \geq \ell_t.$$

Thus,

$$\ell_0 \leq \ell_1 \leq \ell_2 \leq \cdots.$$

So the sequence (ℓ_t) is monotone non-decreasing.

If the log-likelihood is bounded above, then by the monotone convergence theorem for real sequences,

$$\boxed{\ell_t \longrightarrow \ell^*}$$

for some finite value ℓ^* .

Therefore, under these conditions, the observed-data likelihood converges.

Step 5: Relation with stationary points

Suppose that the parameter space is such that:

- the likelihood is bounded above;
- the relevant parameter set is compact, or the sequence $\theta^{(t)}$ remains in a compact subset;

- the likelihood and Q are sufficiently regular (for example, continuous and differentiable where required);
- the M-step finds an exact maximum of $Q(\theta \mid \theta^{(t)})$.

Then the parameter sequence has at least one accumulation point.

Let

$$\theta^*$$

be an accumulation point of the EM sequence.

Because the observed log-likelihood is non-decreasing and converges, the increase in likelihood must vanish asymptotically:

$$\ell(\theta^{(t+1)}) - \ell(\theta^{(t)}) \longrightarrow 0.$$

From

$$\ell(\theta^{(t+1)}) - \ell(\theta^{(t)}) \geq Q(\theta^{(t+1)} \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}),$$

we also obtain that asymptotically there can be no positive gain in the Q function.

Therefore, at an accumulation point θ^* , the M-step can no longer improve Q .

Under the regularity assumptions above, this implies

$$\nabla_{\theta} Q(\theta^* \mid \theta^*) = 0.$$

The EM fixed-point relation then gives

$$\nabla_{\theta} \ell(\theta^*; x) = 0.$$

Hence,

θ^* is a stationary point of the observed-data log-likelihood.

Therefore, the standard convergence statement is:

Under suitable regularity conditions, every accumulation point of the EM sequence is a stationary point of the likelihood.

Important caveat for Gaussian mixtures. For an unrestricted Gaussian mixture model, the observed likelihood is not necessarily bounded above.

For example, a covariance matrix can collapse around a single observation:

$$\Sigma_k \longrightarrow 0.$$

In this situation, the corresponding Gaussian density can become arbitrarily large, and the likelihood can diverge.

Thus, for an unconstrained GMM, one must be careful with the statement “EM converges to the MLE.”

The precise statement is that EM increases the likelihood at every iteration, while convergence to a finite stationary point requires appropriate regularity conditions or constraints on the parameter space.

5.6 Classification EM (CEM)

The standard EM algorithm keeps the memberships

$$\tau_{ik} = P(Z_i = k \mid X_i = x_i)$$

as **soft assignments**.

Thus, an observation can belong partially to several groups.

Classification EM (CEM) introduces a **hard classification step**.

After the E-step, each observation is assigned to the group with the largest posterior probability:

$$\hat{Z}_i = \arg \max_k \tau_{ik}.$$

Equivalently,

$$\tilde{Z}_{ik} = \begin{cases} 1 & \text{if } k = \arg \max_{\ell} \tau_{i\ell}, \\ 0 & \text{otherwise.} \end{cases}$$

The algorithm therefore becomes

$$\boxed{\text{E-step} \longrightarrow \text{Classification step} \longrightarrow \text{M-step.}}$$

In the M-step, the parameters are updated using the hard assignments, for example

$$\hat{p}_k = \frac{n_k}{n},$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i: \hat{Z}_i = k} x_i.$$

Properties of CEM. CEM has the following characteristics:

- It produces a **hard partition** of the observations.
- It is usually more directly focused on classification than standard EM.
- The algorithm can be faster and easier to interpret because every observation is assigned to one group.

- The objective associated with the hard classification is not exactly the same as the ordinary observed-data likelihood of standard EM.
- Therefore, the monotonicity guarantee of standard EM for the observed-data likelihood does not automatically carry over to CEM.

Conceptually,

EM: soft membership	CEM: hard membership.
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5.7 Stochastic EM (SEM)

Stochastic EM (SEM) replaces the deterministic classification step by a random sampling step.

After the E-step, the conditional probabilities

$$\tau_{ik} = P(Z_i = k \mid X_i = x_i, \theta^{(t)})$$

are used to randomly generate a group assignment:

$$Z_i^{(t)} \sim \mathcal{M}\left(1, \tau_{i1}^{(t)}, \dots, \tau_{iK}^{(t)}\right).$$

Thus, an observation is assigned to exactly one group, but the assignment is random.

The algorithm is therefore

E-step $\longrightarrow$ Random sampling of $Z \longrightarrow$ M-step.

Properties of SEM.

- SEM produces **hard classifications**, like CEM.
- The classifications are stochastic rather than deterministic.
- The random component can help the algorithm explore different regions of the parameter space and potentially escape some poor local solutions.
- Unlike standard EM, the observed log-likelihood is not necessarily non-decreasing at every iteration.
- The sequence of parameter estimates is therefore stochastic.

The main distinction is

EM	:	soft deterministic memberships
CEM	:	hard deterministic memberships
SEM	:	hard stochastic memberships.

5.8 k -Means as a Particular Case of a Gaussian Mixture Model

The k -means algorithm can be interpreted as a particular restricted case of Gaussian-mixture clustering.

Consider a Gaussian mixture model with

$$X \mid Z = k \sim \mathcal{N}(\mu_k, \sigma^2 I),$$

where

$$\Sigma_k = \sigma^2 I \quad \text{for every } k,$$

and suppose that the mixing proportions are equal:

$$p_1 = \cdots = p_K = \frac{1}{K}.$$

Thus, all groups have:

- the same spherical covariance matrix;
- the same variance in every direction;
- the same mixing proportion.

For a given observation x_i , the posterior probability of group k is proportional to

$$p_k f_k(x_i).$$

Because all p_k are equal and all covariance matrices are equal, maximising the posterior probability is equivalent to maximising the Gaussian density:

$$\hat{Z}_i = \arg \max_k f_k(x_i).$$

For a spherical Gaussian,

$$f_k(x_i) \propto \exp \left(-\frac{1}{2\sigma^2} \|x_i - \mu_k\|^2 \right).$$

Since the exponential function is increasing,

$$\hat{Z}_i = \arg \min_k \|x_i - \mu_k\|^2.$$

Therefore,

assign each observation to its nearest mean.

This is exactly the assignment step of k -means.

5.8.1 The k -Means Objective

The k -means algorithm minimises the within-cluster sum of squared distances:

$$J(\mu, Z) = \sum_{i=1}^n \|x_i - \mu_{Z_i}\|^2.$$

The two k -means steps are:

Assignment step. Given the current means,

$$Z_i = \arg \min_k \|x_i - \mu_k\|^2.$$

Update step. Given the current assignments,

$$\mu_k = \frac{1}{n_k} \sum_{i: Z_i=k} x_i.$$

These are exactly the same two types of operations that appear in hard Gaussian-mixture estimation:

Assignment	$\longleftrightarrow$	Classification
Mean update	$\longleftrightarrow$	M-step.

5.8.2 Connection Between GMM, CEM and k -Means

The relationship can therefore be summarised as follows.

For a general GMM:

$$X \mid Z = k \sim \mathcal{N}(\mu_k, \Sigma_k),$$

the memberships are soft:

$$\tau_{ik} = P(Z_i = k \mid X_i = x_i).$$

For CEM, these responsibilities are converted into hard assignments:

$$Z_i = \arg \max_k \tau_{ik}.$$

For k -means, we additionally impose spherical and common covariance, equal mixing proportions, and use the nearest-mean assignment:

$$Z_i = \arg \min_k \|x_i - \mu_k\|^2.$$

Thus,

k -means can be viewed as a hard, highly constrained Gaussian-mixture clustering procedure.

More precisely, with

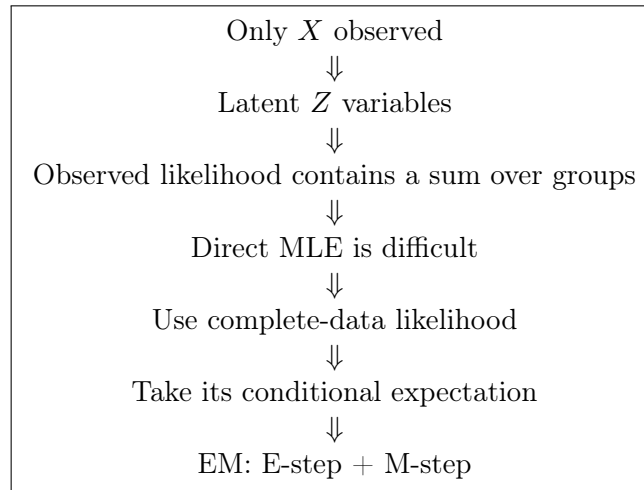
$$\Sigma_k = \sigma^2 I$$

for all k and equal mixing proportions, the Gaussian posterior classification rule becomes nearest-centroid classification.

The variance σ^2 controls the scale of the Gaussian densities. The k -means objective is recovered from the Gaussian log-likelihood up to multiplicative and additive constants when the common variance is fixed.

Summary

The main ideas of this chapter can be summarised by the chain



The three main algorithms are

EM	: soft memberships
CEM	: hard deterministic memberships
SEM	: hard stochastic memberships.

And the main connection with k -means is

k -means $\subset$ hard Gaussian-mixture clustering $\subset$ Gaussian-mixture clustering.
--

The key distinction is that a general GMM learns both the cluster means and covariance structures and uses probabilistic memberships, whereas k -means uses nearest-centroid hard assignments and a common spherical geometry.

6 Proofs of Estimators (1D GMM - Classification)

For $p = 1$, there is only one feature, so we consider a **one-dimensional Gaussian mixture model**:

$$X \mid Z = k \sim \mathcal{N}(\mu_k, \sigma_k^2), \quad k = 1, \dots, K.$$

The one-hot representation of the group variable is

$$\tilde{Z} \sim \mathcal{M}(1, p_1, \dots, p_K),$$

where

$$p_k = P(Z = k), \quad \sum_{k=1}^K p_k = 1.$$

The parameter vector is

$$\theta = (p_1, \dots, p_K, \mu_1, \dots, \mu_K, \sigma_1^2, \dots, \sigma_K^2).$$

For group k , the Gaussian density is

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{1}{2} \frac{(x - \mu_k)^2}{\sigma_k^2}\right).$$

We observe

$$\underline{X} = (X_1, \dots, X_n)$$

and denote the corresponding group variables by

$$\underline{Z} = (Z_1, \dots, Z_n).$$

Using the one-hot representation $\tilde{Z}_{ik}$, where

$$\tilde{Z}_{ik} = \begin{cases} 1 & \text{if } Z_i = k, \\ 0 & \text{otherwise,} \end{cases}$$

the complete-data log-likelihood is

$$\ln L(\underline{X}, \underline{Z}, \theta) = \sum_{i=1}^n \sum_{k=1}^K \tilde{Z}_{ik} \left[\ln p_k - \frac{1}{2} \ln \sigma_k^2 - \ln \sqrt{2\pi} - \frac{1}{2} \frac{(x_i - \mu_k)^2}{\sigma_k^2} \right].$$

We now derive the maximum likelihood estimators of the three types of parameters:

$$p_k, \quad \mu_k, \quad \sigma_k^2.$$

6.1 Estimation of the Mixing Proportions p_k

The mixing proportions satisfy the constraint

$$\sum_{k=1}^K p_k = 1.$$

Equivalently,

$$\sum_{k=1}^K p_k - 1 = 0.$$

Following the notation used by the professor, we can write

$$\sum_{k=1}^K p_k - 1 = \psi(x).$$

To find the extremum of a function $\phi(x)$ under the constraint

$$\psi(x) = 0,$$

we introduce a Lagrange multiplier λ and optimize

$$F(x, \lambda) = \phi(x) + \lambda\psi(x).$$

In our case, we therefore define the Lagrangian

$$F(\theta, \lambda) = \ln L(\underline{X}, Z, \theta) + \lambda \left(1 - \sum_{k=1}^K p_k \right).$$

We use the conditions

$$\frac{\partial F}{\partial p_k} = 0, \quad \frac{\partial F}{\partial \lambda} = 0.$$

When differentiating with respect to p_k , all terms that do not depend on p_k can be ignored. Therefore,

$$F = \sum_{k=1}^K \left(\sum_{i=1}^n \tilde{Z}_{ik} \right) \ln p_k + \lambda \left(1 - \sum_{k=1}^K p_k \right) + \text{terms independent of the } p_k.$$

Define

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik},$$

where n_k is the number of observations belonging to group k .

Thus, the part of the Lagrangian that depends on the mixing proportions becomes

$$F = \sum_{k=1}^K n_k \ln p_k + \lambda \left(1 - \sum_{k=1}^K p_k \right) + \text{constant}.$$

Step 1: Differentiate with respect to p_k . For a fixed group k ,

$$\frac{\partial F}{\partial p_k} = \frac{n_k}{p_k} - \lambda.$$

At the optimum,

$$\frac{n_k}{p_k} - \lambda = 0.$$

Therefore,

$$p_k = \frac{n_k}{\lambda}.$$

Step 2: Differentiate with respect to λ . The derivative with respect to the Lagrange multiplier gives back the constraint:

$$\frac{\partial F}{\partial \lambda} = 1 - \sum_{k=1}^K p_k = 0.$$

Therefore,

$$\sum_{k=1}^K p_k = 1.$$

Substituting

$$p_k = \frac{n_k}{\lambda}$$

into the constraint gives

$$\sum_{k=1}^K \frac{n_k}{\lambda} = 1.$$

Since every observation belongs to exactly one group,

$$\sum_{k=1}^K n_k = n.$$

Therefore,

$$\frac{n}{\lambda} = 1 \quad \implies \quad \lambda = n.$$

Hence,

$$\boxed{\hat{p}_k = \frac{n_k}{n}}.$$

Since

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik},$$

we can equivalently write

$$\hat{p}_k = \frac{1}{n} \sum_{i=1}^n \tilde{Z}_{ik}.$$

Thus, the estimated mixing proportion of group k is simply the proportion of observations belonging to group k .

6.2 Estimation of the Mean μ_k

We now estimate the mean μ_k of group k .

Unlike the mixing proportions, the means are not subject to a constraint. Therefore, we can directly differentiate the log-likelihood with respect to μ_k .

Starting from

$$\ln L(\underline{X}, Z, \theta) = \sum_{i=1}^n \sum_{k=1}^K \tilde{Z}_{ik} \left[\ln p_k - \frac{1}{2} \ln \sigma_k^2 - \ln \sqrt{2\pi} - \frac{1}{2} \frac{(x_i - \mu_k)^2}{\sigma_k^2} \right],$$

we only keep the terms that depend on μ_k :

$$\ln L = -\frac{1}{2} \sum_{i=1}^n \tilde{Z}_{ik} \frac{(x_i - \mu_k)^2}{\sigma_k^2} + \text{constant}.$$

We differentiate with respect to μ_k :

$$\frac{\partial \ln L}{\partial \mu_k} = -\frac{1}{2} \sum_{i=1}^n \tilde{Z}_{ik} \frac{\partial}{\partial \mu_k} \left(\frac{(x_i - \mu_k)^2}{\sigma_k^2} \right).$$

Since σ_k^2 does not depend on μ_k ,

$$\frac{\partial}{\partial \mu_k} (x_i - \mu_k)^2 = -2(x_i - \mu_k).$$

Therefore,

$$\frac{\partial \ln L}{\partial \mu_k} = \frac{1}{\sigma_k^2} \sum_{i=1}^n \tilde{Z}_{ik} (x_i - \mu_k).$$

At the maximum likelihood estimator,

$$\frac{\partial \ln L}{\partial \mu_k} = 0.$$

Thus,

$$\frac{1}{\sigma_k^2} \sum_{i=1}^n \tilde{Z}_{ik} (x_i - \mu_k) = 0.$$

Since $\sigma_k^2 > 0$, we can multiply by σ_k^2 :

$$\sum_{i=1}^n \tilde{Z}_{ik}(x_i - \mu_k) = 0.$$

Expanding the expression,

$$\sum_{i=1}^n \tilde{Z}_{ik}x_i - \mu_k \sum_{i=1}^n \tilde{Z}_{ik} = 0.$$

Using

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik},$$

we obtain

$$\sum_{i=1}^n \tilde{Z}_{ik}x_i - \mu_k n_k = 0.$$

Therefore,

$$\mu_k n_k = \sum_{i=1}^n \tilde{Z}_{ik}x_i$$

and hence

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik}x_i.$$

Thus, the maximum likelihood estimator of μ_k is simply the **sample mean of the observations belonging to group k** .

6.3 Estimation of the Variance σ_k^2

Finally, we estimate the variance σ_k^2 of group k .

Again, there is no constraint on σ_k^2 , so we directly differentiate the log-likelihood with respect to σ_k^2 .

The terms depending on σ_k^2 are

$$\ln L = \sum_{i=1}^n \tilde{Z}_{ik} \left[-\frac{1}{2} \ln \sigma_k^2 - \frac{1}{2} \frac{(x_i - \mu_k)^2}{\sigma_k^2} \right] + \text{constant}.$$

Therefore,

$$\frac{\partial \ln L}{\partial \sigma_k^2} = \sum_{i=1}^n \tilde{Z}_{ik} \frac{\partial}{\partial \sigma_k^2} \left[-\frac{1}{2} \ln \sigma_k^2 - \frac{1}{2} \frac{(x_i - \mu_k)^2}{\sigma_k^2} \right].$$

For the first term,

$$\frac{\partial}{\partial \sigma_k^2} \left(-\frac{1}{2} \ln \sigma_k^2 \right) = -\frac{1}{2\sigma_k^2}.$$

For the second term,

$$\frac{\partial}{\partial \sigma_k^2} \left(-\frac{1}{2} \frac{(x_i - \mu_k)^2}{\sigma_k^2} \right) = \frac{1}{2} \frac{(x_i - \mu_k)^2}{(\sigma_k^2)^2}.$$

Therefore,

$$\frac{\partial \ln L}{\partial \sigma_k^2} = \sum_{i=1}^n \tilde{Z}_{ik} \left[-\frac{1}{2\sigma_k^2} + \frac{1}{2} \frac{(x_i - \mu_k)^2}{(\sigma_k^2)^2} \right].$$

At the maximum likelihood estimator,

$$\frac{\partial \ln L}{\partial \sigma_k^2} = 0.$$

Thus,

$$\sum_{i=1}^n \tilde{Z}_{ik} \left[-\frac{1}{2\sigma_k^2} + \frac{1}{2} \frac{(x_i - \mu_k)^2}{(\sigma_k^2)^2} \right] = 0.$$

Multiplying by $2(\sigma_k^2)^2$ gives

$$\sum_{i=1}^n \tilde{Z}_{ik} [-\sigma_k^2 + (x_i - \mu_k)^2] = 0.$$

Therefore,

$$\sum_{i=1}^n \tilde{Z}_{ik} (x_i - \mu_k)^2 = \sigma_k^2 \sum_{i=1}^n \tilde{Z}_{ik}.$$

Using

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik},$$

we obtain

$$\sum_{i=1}^n \tilde{Z}_{ik} (x_i - \mu_k)^2 = n_k \sigma_k^2.$$

Hence,

$$\hat{\sigma}_k^2 = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} (x_i - \hat{\mu}_k)^2.$$

Thus, the maximum likelihood estimator of σ_k^2 is the **average squared deviation from the estimated group mean**, calculated using only the observations belonging to group k .

Summary of the Three Estimators

For a one-dimensional Gaussian mixture with known group memberships, the maximum likelihood estimators are

$$\hat{p}_k = \frac{n_k}{n}$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} x_i$$

and

$$\hat{\sigma}_k^2 = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} (x_i - \hat{\mu}_k)^2.$$

In other words,

$$\begin{aligned} \hat{p}_k &= \text{proportion of observations in group } k, \\ \hat{\mu}_k &= \text{mean of the observations in group } k, \\ \hat{\sigma}_k^2 &= \text{variance of the observations in group } k. \end{aligned}$$

These are exactly the usual empirical estimators applied separately within each group.

7 Proofs of Estimators (Multivariate GMM - Classification)

We now consider the multivariate case, where each observation has p features.

For $p > 1$, we consider a **multivariate Gaussian mixture model**:

$$X \mid Z = k \sim \mathcal{N}_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad k = 1, \dots, K,$$

where

$$\boldsymbol{\mu}_k \in \mathbb{R}^p$$

is the mean vector of group k and

$$\boldsymbol{\Sigma}_k \in \mathbb{R}^{p \times p}$$

is the covariance matrix of group k .

Since $\boldsymbol{\Sigma}_k$ is a covariance matrix, it is symmetric positive definite:

$$\boldsymbol{\Sigma}_k^\top = \boldsymbol{\Sigma}_k, \quad \boldsymbol{\Sigma}_k \succ 0.$$

The one-hot representation of the group variable is

$$\tilde{Z} \sim \mathcal{M}(1, p_1, \dots, p_K),$$

where

$$p_k = P(Z = k), \quad \sum_{k=1}^K p_k = 1.$$

The parameter vector is now

$$\theta = (p_1, \dots, p_K, \boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K, \boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_K).$$

For group k , the multivariate Gaussian density is

$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}_k|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \right).$$

We observe

$$\underline{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n),$$

where

$$\mathbf{X}_i \in \mathbb{R}^p,$$

and denote the corresponding group variables by

$$\underline{Z} = (Z_1, \dots, Z_n).$$

Using the one-hot representation $\tilde{Z}_{ik}$, where

$$\tilde{Z}_{ik} = \begin{cases} 1 & \text{if } Z_i = k, \\ 0 & \text{otherwise,} \end{cases}$$

the complete-data log-likelihood is

$$\ln L(\underline{X}, Z, \theta) = \sum_{i=1}^n \sum_{k=1}^K \tilde{Z}_{ik} \left[\ln p_k - \frac{p}{2} \ln(2\pi) - \frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k) \right].$$

We now derive the maximum likelihood estimators of the three types of parameters:

$$p_k, \quad \boldsymbol{\mu}_k, \quad \Sigma_k.$$

As in the univariate case, we treat the group memberships $Z_1, \dots, Z_n$ as observed. Therefore, these are the **complete-data maximum likelihood estimators**.

7.1 Estimation of the Mixing Proportions p_k

The mixing proportions satisfy the constraint

$$\sum_{k=1}^K p_k = 1.$$

We therefore introduce a Lagrange multiplier λ and define the Lagrangian

$$F(\theta, \lambda) = \ln L(\underline{X}, Z, \theta) + \lambda \left(1 - \sum_{k=1}^K p_k \right).$$

We use the conditions

$$\frac{\partial F}{\partial p_k} = 0, \quad \frac{\partial F}{\partial \lambda} = 0.$$

Define

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik},$$

where n_k is the number of observations belonging to group k .

Keeping only the terms that depend on the mixing proportions, the Lagrangian becomes

$$F = \sum_{k=1}^K n_k \ln p_k + \lambda \left(1 - \sum_{k=1}^K p_k \right) + \text{constant}.$$

Step 1: Differentiate with respect to p_k . For a fixed group k ,

$$\frac{\partial F}{\partial p_k} = \frac{n_k}{p_k} - \lambda.$$

At the optimum,

$$\frac{n_k}{p_k} - \lambda = 0.$$

Therefore,

$$p_k = \frac{n_k}{\lambda}.$$

Step 2: Differentiate with respect to λ . The derivative with respect to the Lagrange multiplier gives back the constraint:

$$\frac{\partial F}{\partial \lambda} = 1 - \sum_{k=1}^K p_k = 0.$$

Therefore,

$$\sum_{k=1}^K p_k = 1.$$

Substituting

$$p_k = \frac{n_k}{\lambda}$$

into the constraint gives

$$\sum_{k=1}^K \frac{n_k}{\lambda} = 1.$$

Since every observation belongs to exactly one group,

$$\sum_{k=1}^K n_k = n.$$

Therefore,

$$\frac{n}{\lambda} = 1 \quad \implies \quad \lambda = n.$$

Hence,

$$\boxed{\hat{p}_k = \frac{n_k}{n}}.$$

Since

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik},$$

we can equivalently write

$$\boxed{\hat{p}_k = \frac{1}{n} \sum_{i=1}^n \tilde{Z}_{ik}}.$$

Thus, exactly as in the univariate case, the estimated mixing proportion of group k is the **proportion of observations belonging to group k** .

7.2 Estimation of the Mean Vector

We now estimate the mean vector

$$\boldsymbol{\mu}_k \in \mathbb{R}^p$$

of group k .

Unlike the mixing proportions, the mean vectors are not subject to a constraint. Therefore, we directly differentiate the log-likelihood with respect to $\boldsymbol{\mu}_k$.

Keeping only the terms that depend on $\boldsymbol{\mu}_k$ gives

$$\ln L = -\frac{1}{2} \sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k) + \text{constant}.$$

Since $\boldsymbol{\Sigma}_k$ is symmetric positive definite, its inverse is also symmetric:

$$(\boldsymbol{\Sigma}_k^{-1})^\top = \boldsymbol{\Sigma}_k^{-1}.$$

We use the standard matrix derivative

$$\frac{\partial}{\partial \boldsymbol{\mu}} \left[(\mathbf{x} - \boldsymbol{\mu})^\top A (\mathbf{x} - \boldsymbol{\mu}) \right] = -2A(\mathbf{x} - \boldsymbol{\mu}),$$

when A is symmetric.

Therefore,

$$\begin{aligned} \frac{\partial \ln L}{\partial \boldsymbol{\mu}_k} &= -\frac{1}{2} \sum_{i=1}^n \tilde{Z}_{ik} [-2\boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k)] \\ &= \boldsymbol{\Sigma}_k^{-1} \sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k). \end{aligned}$$

At the maximum likelihood estimator,

$$\frac{\partial \ln L}{\partial \boldsymbol{\mu}_k} = \mathbf{0}.$$

Thus,

$$\boldsymbol{\Sigma}_k^{-1} \sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k) = \mathbf{0}.$$

Since $\boldsymbol{\Sigma}_k$ is invertible, we can multiply both sides by $\boldsymbol{\Sigma}_k$:

$$\sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k) = \mathbf{0}.$$

Expanding the expression gives

$$\sum_{i=1}^n \tilde{Z}_{ik} \mathbf{x}_i - \boldsymbol{\mu}_k \sum_{i=1}^n \tilde{Z}_{ik} = \mathbf{0}.$$

Using

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik},$$

we obtain

$$\sum_{i=1}^n \tilde{Z}_{ik} \mathbf{x}_i - n_k \boldsymbol{\mu}_k = \mathbf{0}.$$

Therefore,

$$n_k \boldsymbol{\mu}_k = \sum_{i=1}^n \tilde{Z}_{ik} \mathbf{x}_i$$

and hence

$$\hat{\boldsymbol{\mu}}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} \mathbf{x}_i.$$

Thus, the maximum likelihood estimator of the mean vector is the **sample mean vector of the observations belonging to group k** .

7.3 Estimation of the Covariance Matrix

Finally, we estimate the covariance matrix

$$\boldsymbol{\Sigma}_k.$$

For a fixed group k , the terms of the log-likelihood that depend on $\boldsymbol{\Sigma}_k$ are

$$\ln L = \sum_{i=1}^n \tilde{Z}_{ik} \left[-\frac{1}{2} \ln |\boldsymbol{\Sigma}_k| - \frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k) \right] + \text{constant}.$$

For convenience, define

$$\mathbf{d}_{ik} = \mathbf{x}_i - \boldsymbol{\mu}_k.$$

Then the relevant part of the log-likelihood can be written as

$$\ln L = -\frac{1}{2} \sum_{i=1}^n \tilde{Z}_{ik} \left[\ln |\boldsymbol{\Sigma}_k| + \mathbf{d}_{ik}^\top \boldsymbol{\Sigma}_k^{-1} \mathbf{d}_{ik} \right] + \text{constant}.$$

We use the following standard matrix derivatives:

$$\frac{\partial}{\partial \boldsymbol{\Sigma}} \ln |\boldsymbol{\Sigma}| = \boldsymbol{\Sigma}^{-1},$$

and

$$\frac{\partial}{\partial \mathbf{\Sigma}} \left(\mathbf{d}^\top \mathbf{\Sigma}^{-1} \mathbf{d} \right) = -\mathbf{\Sigma}^{-1} \mathbf{d} \mathbf{d}^\top \mathbf{\Sigma}^{-1},$$

where $\mathbf{\Sigma}$ is symmetric positive definite.

Therefore,

$$\begin{aligned} \frac{\partial \ln L}{\partial \mathbf{\Sigma}_k} &= -\frac{1}{2} \sum_{i=1}^n \tilde{Z}_{ik} \left[\mathbf{\Sigma}_k^{-1} - \mathbf{\Sigma}_k^{-1} \mathbf{d}_{ik} \mathbf{d}_{ik}^\top \mathbf{\Sigma}_k^{-1} \right] \\ &= -\frac{1}{2} \sum_{i=1}^n \tilde{Z}_{ik} \left[\mathbf{\Sigma}_k^{-1} - \mathbf{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top \mathbf{\Sigma}_k^{-1} \right]. \end{aligned}$$

At the maximum likelihood estimator,

$$\frac{\partial \ln L}{\partial \mathbf{\Sigma}_k} = \mathbf{0}.$$

Therefore,

$$\sum_{i=1}^n \tilde{Z}_{ik} \left[\mathbf{\Sigma}_k^{-1} - \mathbf{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top \mathbf{\Sigma}_k^{-1} \right] = \mathbf{0}.$$

Expanding the sum gives

$$\left(\sum_{i=1}^n \tilde{Z}_{ik} \right) \mathbf{\Sigma}_k^{-1} - \mathbf{\Sigma}_k^{-1} \left[\sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top \right] \mathbf{\Sigma}_k^{-1} = \mathbf{0}.$$

Using

$$n_k = \sum_{i=1}^n \tilde{Z}_{ik},$$

we obtain

$$n_k \mathbf{\Sigma}_k^{-1} - \mathbf{\Sigma}_k^{-1} \left[\sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top \right] \mathbf{\Sigma}_k^{-1} = \mathbf{0}.$$

Multiplying on the left and right by $\mathbf{\Sigma}_k$ gives

$$n_k \mathbf{\Sigma}_k - \sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top = \mathbf{0}.$$

Therefore,

$$\sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top = n_k \mathbf{\Sigma}_k.$$

Hence,

$$\mathbf{\Sigma}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top.$$

Since the maximum likelihood estimator of the mean vector is $\hat{\boldsymbol{\mu}}_k$, the covariance estimator is therefore

$$\hat{\boldsymbol{\Sigma}}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^\top.$$

Thus, the maximum likelihood estimator of the covariance matrix is the empirical covariance matrix of the observations belonging to group k .

Notice that, as in the univariate case, the denominator is n_k , not $n_k - 1$. This is because we are deriving the maximum likelihood estimator rather than the unbiased estimator of the covariance matrix.

Summary of the Three Estimators

For a multivariate Gaussian mixture with known group memberships, the maximum likelihood estimators are

$$\hat{p}_k = \frac{n_k}{n}$$

$$\hat{\boldsymbol{\mu}}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} \mathbf{x}_i$$

and

$$\hat{\boldsymbol{\Sigma}}_k = \frac{1}{n_k} \sum_{i=1}^n \tilde{Z}_{ik} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^\top.$$

In other words,

$$\begin{aligned} \hat{p}_k &= \text{proportion of observations in group } k, \\ \hat{\boldsymbol{\mu}}_k &= \text{mean vector of the observations in group } k, \\ \hat{\boldsymbol{\Sigma}}_k &= \text{covariance matrix of the observations in group } k. \end{aligned}$$

These are exactly the multivariate empirical estimators applied separately within each group.

The transition from the univariate to the multivariate case is therefore straightforward:

$$\begin{aligned} p_k &\longrightarrow \hat{p}_k, \\ \mu_k &\longrightarrow \hat{\boldsymbol{\mu}}_k, \\ \sigma_k^2 &\longrightarrow \hat{\boldsymbol{\Sigma}}_k. \end{aligned}$$

The scalar squared deviation

$$(x_i - \mu_k)^2$$

is replaced by the outer product

$$(\boldsymbol{x}_i - \boldsymbol{\mu}_k)(\boldsymbol{x}_i - \boldsymbol{\mu}_k)^\top,$$

which produces the $p \times p$ covariance matrix.